

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Systems for Data Science, Performance Engineering, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

JavaScript, TypeScript, Python, Java, C++, SQL, Bash, HTML/CSS

Backend & APIs

Node.js, Express, FastAPI, Fastify, REST APIs, async services, integration testing, authentication, rate limiting, audit logging

Web & Full-Stack

React, Electron, Webpack, GraphQL

Databases & DevOps

MongoDB, PostgreSQL, Redis, SQLite, Docker, Git, GitHub Actions, CI/CD, Linux

AI-Assisted Dev & GenAI

Claude / Claude Code, GitHub Copilot, Cursor, ChatGPT / Codex, OpenAI / Groq APIs, RAG, LLM agents, PyTorch, Hugging-Face Transformers

Experience

Wysa

July 2023 – July 2025

Backend Engineer

Bengaluru, India

- Engineered **multi-tenant NHS eTriage platform** for **8+ UK clients** (Node.js, MongoDB) processing **10,000+ monthly submissions** as a production-grade system integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines, contributing to **\$340K+** in revenue.
- Architected full-stack data annotation platform (**React, Node.js, MongoDB**) used daily by AI team to label data for production models, replacing spreadsheet workflows with secure role-based REST APIs; cut manual operations by **100+ hours/month** and improved labeling throughput by **60%**.
- Owned backend reliability across the eTriage stack: monitored production logs during feature releases and client integrations, shipped fixes for tenant-specific incidents, hardened authentication and rate limiting, added integration tests for regression protection, and remediated **20+ Vulnerability Assessment and Penetration Testing (VAPT) findings** to protect sensitive clinical data.

Cario Growth Services Pvt. Ltd.

November 2022 – May 2023

Backend Developer Intern

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify, backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core codebase from **JavaScript to TypeScript** across multiple repositories and prototyped LLM-powered context-aware replies in a chat product, improving type safety, maintainability, and end-user response quality.

Projects

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Developed multi-stage agentic web-discovery service that plans typed retrieval, reranks pages, classifies evidence regimes, extracts entities with deterministic parsers before LLM fallback, verifies outputs, and returns cell-level provenance for every result.
- Hardened service with **150+ automated tests**, provider routing and fallback across LLM backends, pipeline instrumentation, and reviewer-facing trust user interface (UI) for inspecting extraction quality and ranking transparency.

Sonare - Offline Sign ↔ Speech Cross-Platform Desktop App

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Built **cross-platform desktop application** integrating camera, microphone, frontend UI, and backend services into on-device pipeline using MediaPipe hand tracking, gesture classification, whisper.cpp Automatic Speech Recognition (ASR), and local Text-To-Speech (TTS) — end-to-end product prototyped, designed, and shipped under hackathon time pressure.
- Engineered **FastAPI backend** and **Electron/React frontend** to run fully offline with no external network dependencies; designed async APIs and queues to keep UI responsive while handling video and audio streams locally on device.

Spark ETL Performance Optimization - NYC TLC Taxi Dataset

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark, PySpark, Databricks, SQL, Python

- Optimized a Spark ETL pipeline on the NYC TLC taxi dataset with broadcast-join tuning, caching, partitioning, adaptive Spark settings, column pruning, and precomputed fields to reduce shuffle volume and wall-clock runtime.
- Used **Spark UI** stage-level metrics to localize shuffle and skew bottlenecks and document a repeatable optimization workflow comparing naive vs. tuned runs across cluster resource usage.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023** — applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.